

A global lower bound for every endpoint odds coordinate

Asymptotic research note; independently reviewed

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Let $p_1, \dots, p_m : [K] \rightarrow [0, 1]$ be nondecreasing, with positive row weights w_q summing to one, and suppose

$$\mathbb{E}_w \prod_{j \in S} p_j = \frac{1}{|S| + 1} \quad (S \subseteq [m]).$$

On rows where $A = \prod_j p_j > 0$, write $r_j = (1 - p_j)/p_j$ and $\lambda = \sum_j r_j$.

Theorem 1. *For $m \geq 2$, every row q with $A(q) > 0$ and $L = \lambda(q) \geq 32$ satisfies*

$$r_i(q) \geq \frac{\min\{L, m - 1\}}{32m} \quad (1 \leq i \leq m). \quad (1)$$

Proof. Fix i . If $r_i(q) > L/2$, the assertion is immediate. Otherwise put $\ell = \lfloor \min\{L, m - 1\}/16 \rfloor$ and choose J to consist of the ℓ smallest odds among the $m - 1$ indices other than i , evaluated at row q . Then

$$\lambda_{-(J \cup \{i\})}(q) \geq \left(1 - \frac{\ell}{m - 1}\right) (L - r_i(q)) \geq \frac{15L}{32}. \quad (2)$$

For a failure set T , define the probability measure

$$\nu_T = (m + 1) \binom{m}{|T|} w \prod_{j \in T} (1 - p_j) \prod_{j \notin T} p_j.$$

The mixed moments imply this normalization. They also imply

$$\mathbb{E}_{\nu_T} \lambda_{-T} \leq |T| + 1. \quad (3)$$

Indeed for each $j \notin T$ the expectation of r_j is at most $(|T| + 1)/(m - |T|)$, by the ratio of the probabilities of the specified failure patterns $T \cup \{j\}$ and

T . The inequality permits zero p_j rows, whose mass is counted only by the latter pattern.

Use (3) for $T = J \cup \{i\}$. By monotonicity and (2), every row strictly before q in the support of ν_T has remaining odds sum at least $15L/32$. As $\ell + 2 \leq L/16 + 2 \leq L/8$, Markov's inequality gives

$$\nu_T(\{s < q\}) \leq \frac{\ell + 2}{15L/32} \leq \frac{4}{15}.$$

In particular the suffix $s \geq q$ has ν_T mass at least $1/2$. All p_j are positive on that suffix. Put $d = m - \ell$; there

$$\nu_T = \frac{d}{\ell + 1} r_i \nu_J.$$

Since $r_i(s) \leq r_i(q)$ on the suffix, we obtain

$$\frac{1}{2} \leq \nu_T(\{s \geq q\}) \leq \frac{d}{\ell + 1} r_i(q) \nu_J(\{s \geq q\}) \leq \frac{d}{\ell + 1} r_i(q).$$

Now $d \leq m$ and $\ell + 1 > \min\{L, m - 1\}/16$, proving (1). $\square$

Corollary 1 (Rows with zero success product). *For $m \geq 2$, if $A(q) = 0$, then every column at that row satisfies*

$$p_i(q) \leq 32/33.$$

Proof. The claim is immediate if $p_i(q) = 0$. Otherwise order the other odds using the value $+\infty$ for zero entries, put $\ell = \lfloor (m - 1)/16 \rfloor$, and let J be the ℓ smallest of them. Since at least one of the other columns is zero and $\ell < m - 1$, at least one zero remains outside $T = J \cup \{i\}$. Monotonicity implies that this column vanishes on every row up to q . Consequently ν_T gives all its mass to the strict suffix $s > q$. On that suffix $p_i > 0$, so the identity $\nu_T = (m - \ell)r_i \nu_J / (\ell + 1)$ is valid even if other columns vanish: both sides then vanish. Monotonicity of r_i gives

$$1 \leq \frac{m - \ell}{\ell + 1} r_i(q), \quad r_i(q) \geq \frac{\ell + 1}{m - \ell} > \frac{m - 1}{16m} \geq \frac{1}{32}.$$

This is equivalent to the assertion. $\square$

The argument uses a number of conditioned failures that grows with the row's odds sum, and chooses those failures among the smallest odds. It does not require an upper balance theorem and remains valid for $\lambda \gg m$. This is a structural comparison-tensor theorem, not by itself a new lower bound on the number of rows.